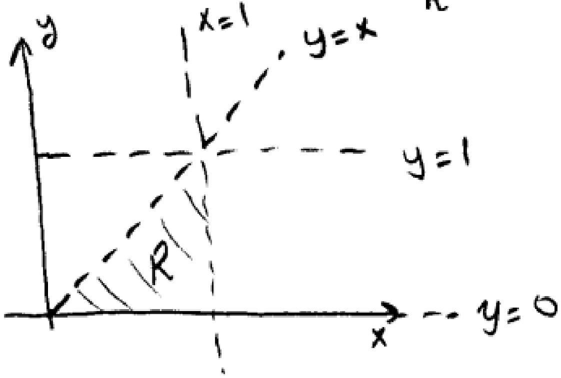
3) $\int_0^1 \int_y^1 \sin(x^2)\,dx\,dy$ integralini, integralin sırasını değiştirerek hesaplayınız.

Question 5 (10 points)

a)Evaluate $\int_0^1 \int_y^1 \sin(x^2)\,dx\,dy = \iint_R \sin(x^2)\,dA = \int_0^1 \int_0^x \sin(x^2)\,dy\,dx$



$$= \int_0^1 y \sin(x^2) \Big|_0^x dx$$

$$= \int_0^1 x \sin(x^2)\,dx$$

$u = x^2 \qquad x=0 \Rightarrow u=0$

$\frac{du}{dx} = 2x \qquad x=1 \Rightarrow u=1$

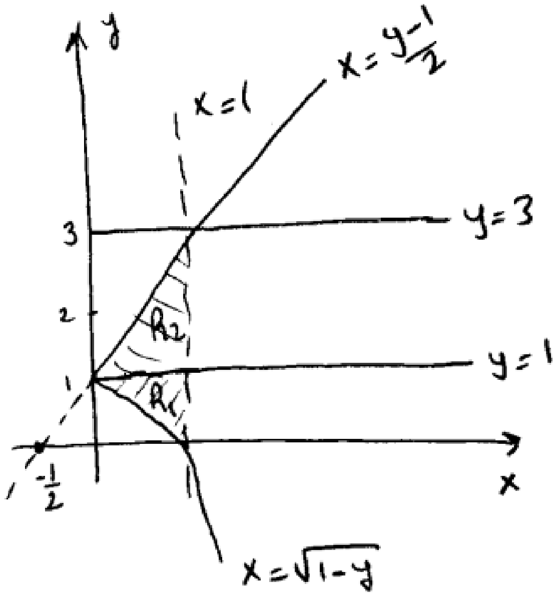
$$= \frac{1}{2} \int_0^1 \sin u\,du = -\frac{1}{2} \cos u \Big|_0^1$$

$$= -\frac{1}{2} (\cos 1 - \cos 0) = \frac{1}{2} (1 - \cos 1)$$

b) $\displaystyle\int_0^1 \int_{\sqrt{1-y}}^1 f(x,y)dxdy \;+\; \int_1^3 \int_{\frac{y-1}{2}}^1 f(x,y)dxdy$

integralleri için, integral bölgesini çiziz. Tek bir integral şeklinde yazınız.

b) After drawing the region for each integral, express $\displaystyle\int_0^1 \int_{\sqrt{1-y}}^1 f(x,y)dxdy$ + $\displaystyle\int_1^3 \int_{\frac{y-1}{2}}^1 f(x,y)dxdy$ as one double integral.



$$= \iint_{R_1} f(x,y)\,dx\,dy + \iint_{R_2} f(x,y)\,dx\,dy$$

$$= \int_0^1 \int_{1-x^2}^{2x+1} f(x,y)\,dy\,dx$$

$x = \sqrt{1-y} \Rightarrow x^2 = 1-y \Rightarrow y = 1-x^2$

$x = \frac{y-1}{2} \Rightarrow 2x+1 = y$

C: $y = x,\ x = 0,\ x^2 + y^2 = 4$ eğrileriyle sınırlı bölgenin, saat yönünün tersine yönlenmiş çevresi olmak üzere

$$\oint_C (-10y^3x^2 + \cos(x^2))\,dx \ + \ (3x^5 + 15y^4x + \sin(y^2))\,dy$$

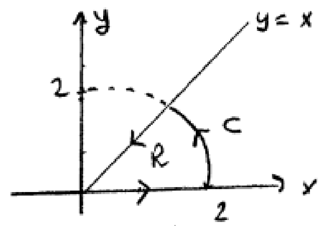
çizgi integralini Green teoremi kullanarak hesaplayınız.

Question 1 (10 points) Evaluate $\oint_C (-10y^3x^2 + \cos(x^2))\,dx \ + \ (3x^5 + 15y^4x + \sin(y^2))\,dy$ where C is counterclockwise oriented perimeter of the region bounded by $y = x,\ x = 0,\ x^2 + y^2 = 4$ in the first quadrant.

Let $P(x,y) = -10y^3x^2 + \cos x^2$, $Q(x,y) = 3x^5 + 15y^4x + \sin y^2$ $\Big\}$ $P_y = -30y^2x^2$, $Q_x = 15x^4 + 15y^4$

- since $P_y \neq Q_x$, (P,Q) is not exact (ie, $\nexists\, f(x,y)$ with $\nabla f = (P,Q)$)
- P & Q are polynomials, hence have all partial derivatives of all order.

Let $C = \partial R$ (boundary curve of the region R)



We can use Green's Theorem;

$$\oint_{C=\partial R} P\,dx + Q\,dy = \iint_R (Q_x - P_y)\,dA$$

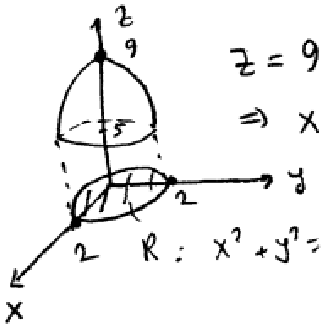
$$= \iint_R 15x^4 + 15y^4 + 30x^2y^2\,dA = 15\iint_R (x^2+y^2)^2\,dA$$

$$= 15\int_0^{\pi/4}\int_0^2 r^4 \cdot r\,dr\,d\theta = 15\int_0^{\pi/4} \frac{r^6}{6}\Big|_0^2 d\theta = \frac{15}{6}\,2^6 \cdot \frac{\pi}{4}$$

$$= 40\pi$$

6) $z = 9 - x^2 - y^2$ paraboloidi ve $z = 5.$ düzlemi ile sınırlı katı cismin hacmini kutupsal koordinatlar kullanarak hesaplayınız.

Question 4 (10 points) Find the volume of the solid bounded by the paraboloid $z = 9 - x^2 - y^2$ and the plane $z = 5$.



$R : x^2 + y^2 = 4$

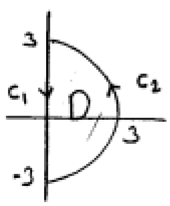
$$z = 9 - x^2 - y^2 = 5$$
$$\Rightarrow x^2 + y^2 = 4$$

$$V = \iint_R (9 - x^2 - y^2 - 5)\, dA$$
$$= \iint_R (4 - r^2)\, r\, dr\, d\theta$$
$$= \int_0^{2\pi} \int_0^2 (4r - r^3)\, dr\, d\theta$$
$$= 2\pi \left[4\frac{r^2}{2} - \frac{r^4}{4} \right]_0^2 = 8\pi$$

7) $f(x,y) = x^2 + y^2 - 2x + 4y + 6$ fonksiyonunun $D = \{(x,y);\ x^2 + y^2 \leq 9,\ \ x \geq 0\}$. bölgesi üzerindeki maksimum ve minimumunu bulunuz.

Surname: .... Name: .... Id: .... Section: ....

Question 6 (12 points) Find the absolute maximum and the absolute minimum of the function $f(x,y) = x^2 + y^2 - 2x + 4y + 6$ on the domain $D = \{(x,y);\ x^2 + y^2 \leq 9,\ \ x \geq 0\}$.



Note that f is a polynomial and D is closed, hence f has absolute Max / min on D. Being a polynomial, f has no singular points. Hence to find abs. Max/min. of f on D, we must check the values of f at critical points inside D and check f on C_1 & on C_2.

inside D • $\left.\begin{array}{l} f_x = 2x - 2 \\ f_y = 2y + 4 \end{array}\right\} = 0 \Rightarrow (x,y) = (1,-2)$ is the only critical point inside.

and $f(1,-2) = 1 + 4 - 2 - 8 + 6 = 1$

• on C_1 : $(x,y) = (0,t)$, $t \in [-3,+3]$ (orientation is not important!)

$\Rightarrow f|_{C_1} = t^2 + 4t + 6$; $t \in [-3,+3]$

$f' = 2t + 4 = 0 \Rightarrow t = -2$ critical pt. : $f|_{C_1}(-2) = 2$

end pts: $f|_{C_1}(-3) = 3$, $f|_{C_1}(3) = 27$

• On C2 : $\left[\begin{array}{ll} \text{Maximize} & f(x,y) = x^2 + y^2 - 2x + 4y + 6 \\ \text{subject to} & g(x,y) = x^2 + y^2 - 9 = 0 \end{array}\right.$

ie, $\left[\begin{array}{ll} \text{Maximize} & f(x,y) = 9 - 2x + 4y + 6 \\ \text{subject to} & g(x,y) = x^2 + y^2 - 9 = 0 \end{array}\right.$

set $\nabla f = \lambda \nabla g \Rightarrow \left.\begin{array}{l} -2 = 2\lambda x \\ 4 = 2\lambda y \\ x^2 + y^2 = 9 \end{array}\right\}$ $x^2 + y^2 = \frac{1}{\lambda^2} + \frac{4}{\lambda^2} = \frac{5}{\lambda^2} = 9 \Rightarrow \lambda = \mp \frac{\sqrt{5}}{3}$

$\Rightarrow x = \frac{-1}{\lambda}$ and $x \geq 0$ so $\lambda < 0$ $\Rightarrow x = 3/\sqrt{5}$

$y = 2/\lambda$ $\Rightarrow y = -\frac{6}{\sqrt{5}}$

$f\left(\frac{3}{\sqrt{5}}, \frac{-6}{\sqrt{5}}\right) = \frac{9}{5} + \frac{36}{5} - \frac{6}{\sqrt{5}} - \frac{24}{\sqrt{5}} + 6 = 15 - \frac{30}{\sqrt{5}}$

Abs. Max $= f(0,3) = 27$, Abs. min $= f(1,-2) = 1$

9) $f(x,y,z) = x^2 + yz - 5$ fonksiyonunun, $x^2+y^2+z^2=1$ küresi üzerindeki maksimum ve minimumunu bulunuz. (Lagrange çarpanları yöntemi kullanınız.)

Question 4 (10 points) Find the MAXIMUM and the minimum values of $f(x,y,z) = x^2+yz-5$ on the sphere $x^2+y^2+z^2=1$.

Maximize/minimize $f(x,y,z) = x^2+yz-5$

subject to $g(x,y,z) = x^2+y^2+z^2-1=0$

Use Lagrange Multipliers Method;

set $\nabla f = \lambda \nabla g$

$$\left.\begin{aligned} f_x &= \lambda g_x \\ f_y &= \lambda g_y \\ f_z &= \lambda g_z \end{aligned}\right\} \quad \left.\begin{aligned} 2x &\overset{1}{=} 2\lambda x \\ z &\overset{2}{=} 2\lambda y \\ y &\overset{3}{=} 2\lambda z \\ x^2+y^2+z^2-1 &\overset{4}{=} 0 \end{aligned}\right] \quad \begin{aligned} &\text{4 equations} \\ &\text{4 unknowns } \checkmark \end{aligned}$$

$1 \Rightarrow 2x(1-\lambda) = 0 \rightarrow x=0$ ($\Rightarrow$ 2 & 4 $\Rightarrow y\neq 0$ & $z \neq 0$)

$\overset{2}{\Rightarrow} \lambda = \dfrac{z}{2y}$ & $\overset{3}{\Rightarrow} \lambda = \dfrac{y}{2z}$

$\Rightarrow \dfrac{z}{2y} = \dfrac{y}{2z} \Rightarrow y^2 = z^2 \Rightarrow y = \mp z$

$\overset{4}{\Rightarrow} 2y^2 = 1 \Rightarrow y = \mp \dfrac{1}{\sqrt{2}}$

$\Rightarrow (x,y,z) = \left(0, \dfrac{1}{\sqrt{2}}, \mp\dfrac{1}{\sqrt{2}}\right)$ or $\left(0, \dfrac{-1}{\sqrt{2}}, \mp\dfrac{1}{\sqrt{2}}\right)$

$x \neq 0 \Rightarrow \lambda = 1 \overset{2}{\Rightarrow} z = 2y \overset{3}{=} 4z \Rightarrow z = 0 = y \Rightarrow x = \mp 1$

$\Rightarrow (x,y,z) = (\mp 1, 0, 0)$

$f\left(0, \dfrac{1}{\sqrt{2}}, \dfrac{-1}{\sqrt{2}}\right) = f\left(0, \dfrac{-1}{\sqrt{2}}, \dfrac{1}{\sqrt{2}}\right) = \dfrac{-1}{2} - 5 = -\dfrac{11}{2}$

$f\left(0, \dfrac{1}{\sqrt{2}}, \dfrac{1}{\sqrt{2}}\right) = f\left(0, \dfrac{-1}{\sqrt{2}}, \dfrac{-1}{\sqrt{2}}\right) = \dfrac{1}{2} - 5 = -\dfrac{9}{2}$

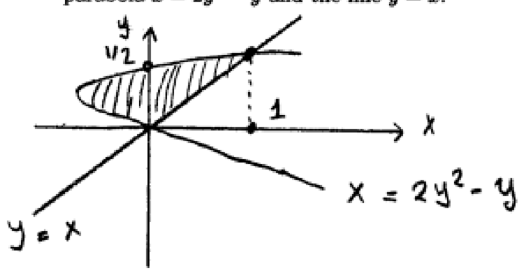
$f(\mp 1, 0, 0) = 1 - 5 = -4$

$\boxed{\text{MAX} = -4}$ $\boxed{\min = -\dfrac{11}{2}}$

10) R bölgesi $x = 2y^2 - y$ parabolü ve $y = x$ doğrusu ile sınırlı bölge olmak üzere $\displaystyle\int_R\int 2xy\ dA$ integralini hesaplayınız.

Question 6 (5+5=10 points)

a)Evaluate $\displaystyle\int_R\int 2xy\ dA$ where R is the finite region in the xy-plane bounded by the parabola $x = 2y^2 - y$ and the line $y = x$.



$$y = 2y^2 - y \Rightarrow 2y^2 - 2y = 0$$
$$2y(y-1) = 0$$
$$y = 0,\ y = 1$$

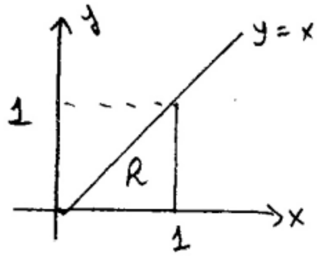
$$\int_0^1 \int_{2y^2-y}^{y} 2xy\,dx\,dy = \int_0^1 x^2 y \Big|_{2y^2-y}^{y} dy = \int_0^1 [y^2 - (2y^2-y)^2]\,y\,dy$$

$$= \int_0^1 (-4y^4 + 4y^3)\,y\,dy = \int_0^1 (-4y^5 + 4y^4)\,dy$$

$$= -4\frac{y^6}{6} + 4\frac{y^5}{5}\Big|_0^1 = -\frac{2}{3} + \frac{4}{5} = \frac{2}{15}$$

11) R bölgesi, köşeleri $(0,0)$, $(1,0)$, $(1,1)$ olan üçgensel bölge olmak üzere $\displaystyle\int_R\!\!\int \frac{xy}{1+x^4}\,dA$ integralini hesaplayınız.

Question 5 (10 points) Evaluate $\displaystyle\int_R\!\!\int \frac{xy}{1+x^4}\,dA$ where R is the triangle with vertices $(0,0)$, $(1,0)$, $(1,1)$.



$$= \int_0^1 \int_0^x \frac{x}{1+x^4}\, y \; dy\, dx = \int_0^1 \frac{x}{1+x^4} \left.\frac{y^2}{2}\right|_{y=0}^{y=x} dx$$

$$= \frac{1}{2}\int_0^1 \frac{x^3}{1+x^4}\, dx = \frac{1}{2}\,\frac{1}{4}\int_1^2 \frac{1}{u}\, du = \frac{1}{8} \ln|u| \Big|_1^2$$

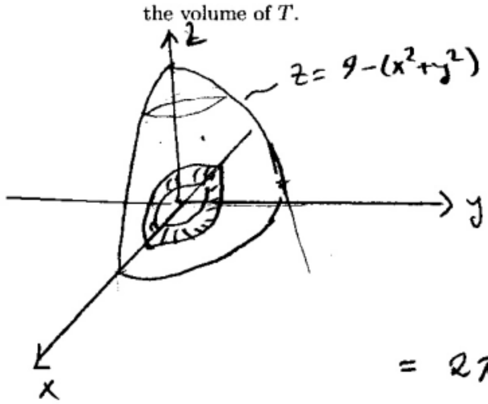
$u = 1 + x^4$

$du = 4x^3\, dx$

$$= \frac{1}{8}\ln 2$$

12) $x^2+y^2=1,\ \ x^2+y^2=4$ silindirleri arasında kalan ve $z=0,\ \ z=9-(x^2+y^2)$ yüzeyleri ile sınırlı katı cismin hacmini polar koordinatlar kullanarak hesaplayınız.

Question 6 (10 points) Consider the solid T that lies between two cylinders $x^2+y^2=1,\ \ x^2+y^2=4$ and bounded by the surfaces $z=0,\ \ z=9-(x^2+y^2)$. Compute the volume of T.



$$V = \iint 9-(x^2+y^2)\, dy\, dx$$

use Polar Coord's

$$= \int_0^{2\pi}\int_1^2 (9-r^2)\, r\, dr\, d\theta$$

$$= 2\pi \left(\frac{9r^2}{2}-\frac{r^4}{4}\right)\Big|_1^2 = 2\pi\left((18-4)-\left(\frac{9}{2}-\frac{1}{4}\right)\right)$$

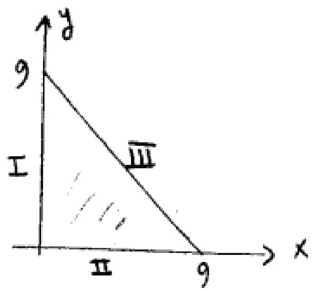
$$= 2\pi\left(14-\frac{17}{4}\right) = \frac{39}{2}\pi$$

13) $f(x,y) = 2 + 2x + 2y - x^2 - y^2$

a)

b) fonksiyonunun, I. bölgede $x = 0,\ \ y = 0,\ \ y = 9 - x.$ doğrularıyla çevrili üçgensel bölge üzerindeki maksimum ve minimumunu hesaplayınız.

b)Find the absolute (global) maximum and absolute minimum values of f on the triangular plate in the first quadrant bounded by the lines $x = 0,\ \ y = 0,\ \ y = 9 - x$.



1-) interior points;

from part (a), (1,1) is the only critical point and $\boxed{f(1,1) = 4}$

2-) boundary points

(i) points on I : $x = 0,\ y \in [0,9]$

$f = 2 + 2y - y^2$ on $[0,9]$

$f' = 2 - 2y = 0 \Rightarrow y = 1$ critical point

$\boxed{f(0,1) = 3}$ $\underbrace{\boxed{f(0,0) = 2} \quad \boxed{f(0,9) = -61}}_{\text{boundary points}}$

(ii) Points on II : $y = 0,\ x \in [0,9]$

$f = 2 + 2x - x^2 \ ;\ x \in [0,9]$

$f' = 2 - 2y = 0 \Rightarrow x = 1$ critical point

$\boxed{f(1,0) = 3}$ $\boxed{f(0,0) = 2}$ $\boxed{f(9,0) = -61}$

(iii) Points on III : $y = 9 - x,\ \ x \in [0,9]$

$\Rightarrow f = 2 + 2x + 2(9-x) - x^2 - (9-x)^2$

$= -61 + 18x - 2x^2 \Rightarrow f' = 18 - 4x = 0 \rightarrow x = 9/2$ critical point

$\boxed{f(\tfrac{9}{2}, \tfrac{9}{2}) = -\tfrac{41}{2}}$ $\boxed{f(0,9) = -61}$ $\boxed{f(9,0) = -61}$

Thus Absolute MAX is $f(1,1) = 4$

Absolute min is $f(9,0) = f(0,9) = -61$

14) $f(x,y,z) = x + y + z$ fonksiyonunun $x^2 + 4y^2 + 9z^2 = 1764$ elipsoidi üzerindeki maksimum ve minimumunu bulunuz.

Question 4 (14 points) Find the maximum and minimum values of the function $f(x,y,z) = x + y + z$ on the ellipsoid $x^2 + 4y^2 + 9z^2 = 1764$.

Use Lagrange Multipliers Method to find Max/min of $f(x,y,z) = x+y+z$
subject to $g(x,y,z) = x^2+4y^2+9z^2 = 17$

set $\nabla f = \lambda \nabla g$

$$\left.\begin{aligned} 1 &\overset{(1)}{=} 2\lambda x \\ 1 &\overset{(2)}{=} 8\lambda y \\ 1 &\overset{(3)}{=} 18\lambda z \\ x^2+4y^2+9z^2 &\overset{(4)}{=} 1764 \end{aligned}\right\} \begin{array}{l} \text{4 equations} \\ \text{4 unknowns} \end{array}$$

$(1) \Rightarrow \lambda \neq 0$

$(1,2,3) \Rightarrow x = \frac{1}{2\lambda}, \quad y = \frac{1}{8\lambda}, \quad z = \frac{1}{18\lambda}$

$(4) \Rightarrow \frac{1}{4\lambda^2} + \frac{1}{16\lambda^2} + \frac{1}{36\lambda^2} = 1764$

$\Rightarrow \frac{49}{144\lambda^2} = 1764 = 42^2$

$\Rightarrow \lambda = \mp \frac{1}{72}$

$\Rightarrow x = \mp 36, \quad y = \mp 9, \quad z = \mp 4$

$\Rightarrow$ Max of $f(x,y,z)$ is $f(36, 9, 4) = 49$

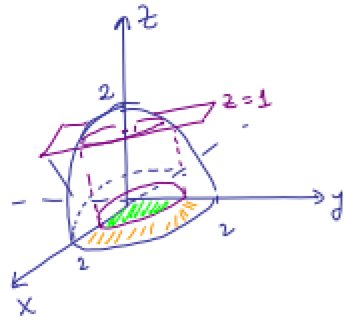
min of $f(x,y,z)$ is $f(-36,-9,-4) = -49$

15) $x^2 + y^2 + z^2 = 4$ küresinin içinde bulunan ve $z=0$ ile $z=1$ düzlemleri arasında kalan katı cismin hacmini

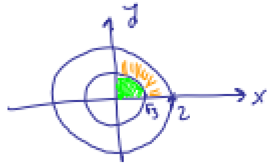
a) kartezyen koordinatlarla
b) silindirik koordinatlarla
c) küresel koordinatlarla hesaplayınız.

Question 2 (24 points) Write (do **not** evaluate) a triple integral to compute the volume of the region that lies inside the sphere $x^2 + y^2 + z^2 = 4$ and between the planes $z = 0$ and $z = 1$ using:

a) Cartesian (rectangular) coordinates.



$$V = \frac{\frac{4}{3}\pi 2^3}{2} - 4\int_0^{\sqrt{3}}\int_0^{\sqrt{3-x^2}}\int_1^{\sqrt{4-x^2-y^2}} 1\, dz\, dy\, dx$$



b) Cylindrical coordinates.

$$V = 4\int_0^{\pi/2}\int_0^{\sqrt{3}}\int_0^{1} 1\ r\ dz\, dr\, d\theta + 4\int_0^{\pi/2}\int_{\sqrt{3}}^{2}\int_0^{\sqrt{4-r^2}} 1 \cdot r\, dz\, dr\, d\theta$$

OR

$$V = \frac{\frac{4}{3}\pi 2^3}{2} - 4\int_0^{\pi/2}\int_0^{\sqrt{3}}\int_1^{\sqrt{4-r^2}} 1\ r\, dz\, dr\, d\theta$$

$x = \rho\sin\phi\cos\theta$
$y = \rho\sin\phi\sin\theta$ $\Big\}\ x^2+y^2 = \rho^2\sin^2\phi$
$z = \rho\cos\phi = 1$

c) Spherical coordinates.

$$V = 4\int_0^{\pi/2}\int_0^{\pi/3}\int_0^{\frac{1}{\cos\phi}} 1\ \rho^2\sin\phi\, d\rho\, d\phi\, d\theta$$

$$+\ 4\int_0^{\pi/2}\int_{\pi/3}^{\pi/2}\int_0^{2} 1\, \rho^2\sin\phi\, d\rho\, d\phi\, d\theta$$

OR

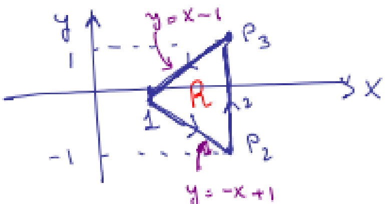
$$V = \frac{\frac{4}{3}\pi 2^3}{2} - 4\int_0^{\pi/2}\int_0^{\pi/3}\int_{1/\cos\phi}^{2} 1\, \rho^2\sin\phi\, d\rho\, d\phi\, d\theta$$

16) C: köşeleri $P_1(1,0),\ P_2(2,-1),\ P_3(2,1)$. olan üçgenin kenarlarının oluşturduğu saat yönünün tersine yönlenmiş eğri olmak üzere

$$\oint_C (y+\ln x)dx + (x^3+e^{-y})dy$$

çizgi integralini Green teoremi kullanarak hesaplayınız.

Question 1 (10 points) Use **Green's theorem** to evaluate the line integral $\oint_C (y+\ln x)dx + (x^3+e^{-y})dy$ where C is the positively oriented boundary of the triangle with vertices $P_1(1,0),\ P_2(2,-1),\ P_3(2,1)$.



$\oint_C M dx + N dy$ where $M(x,y) = y + \ln x$
$N(x,y) = x^3 + e^{-y}$

Note that $N_x - M_y = 3x^2 - 1$

M & N are differentiable everywhere, C is positively oriented,

hence by Green's thm, we have

$$\oint_{C=\partial R} M dx + N dy = \iint_R N_x - M_y \, dA = \int_1^2 \int_{-x+1}^{x-1} 3x^2 - 1 \, dy \, dx = \int_1^2 (3x^2-1)\underbrace{\left[(x-1) - (-x+1)\right]}_{2x-2} dx$$

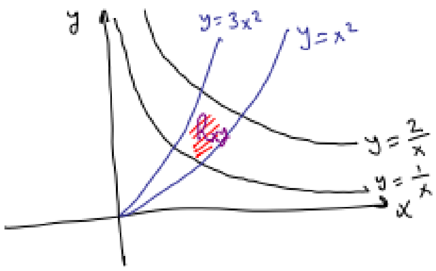
$$= 2\int_1^2 3x^3 - 3x^2 - x + 1 \, dx = 2\left[\frac{3x^4}{4} - \frac{3x^3}{3} - \frac{x^2}{2} + x\right]_1^2$$

$$= 2\left[\left(3\cdot 4 - 8 - \cancel{2} + \cancel{2}\right) - \left(\frac{3}{4} - \cancel{1} - \frac{1}{2} + \cancel{1}\right)\right] = 2\left[4 - \frac{1}{4}\right]$$

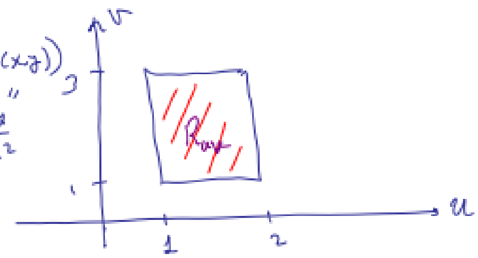
$$= 2\,\frac{15}{4} = 15/2$$

17) $y=\frac{1}{x},\ y=\frac{2}{x},\ y=x^2$ and $y=3x^2$. eğrileriyle sınırlı bölgenin alanını değişken değiştirerek bulunuz. (İpucu $u=xy$, $v=\frac{y}{x^2}$)

Question 5 (10 points) Find the area of the region in the first quadrant bounded by the curves $y=\frac{1}{x}$, $y=\frac{2}{x}$, $y=x^2$ and $y=3x^2$.



$r(x,y)=\left(u(x,y),\ v(x,y)\right)$, $u(x,y) = xy$, $v(x,y) = \frac{y}{x^2}$



$$\text{Area}=\iint_{R_{xy}} 1\,dy\,dx=\iint_{R_{uv}} 1\left|\frac{\partial(x,y)}{\partial(u,v)}\right| dv\,du=\int_1^2\int_1^3\left|\frac{1}{3v}\right|dv\,du$$

$$\frac{\partial(x,y)}{\partial(u,v)}=\frac{1}{\frac{\partial(u,v)}{\partial(x,y)}}=\frac{1}{\det\begin{bmatrix}u_x & u_y\\ v_x & v_y\end{bmatrix}}=\frac{1}{\det\begin{bmatrix}y & x\\ \frac{-2y}{x^3} & \frac{1}{x^2}\end{bmatrix}}=\frac{1}{\frac{y}{x^2}+\frac{2\cancel{x}y}{x^{\cancel{3}2}}}=\frac{1}{\frac{3y}{x^2}}=\frac{x^2}{3y}=\frac{1}{3v}$$

$$=\frac{1}{3}\ln v\Big|_1^3=\frac{1}{3}\left(\ln(3)-\cancel{\ln(1)}^{\,0}\right)=\frac{\ln 3}{3}$$

18) $x^3 + y^3 + 3x^2y = 15.$ eşitliğini sağlayan, negatif olmayan x sayılarının toplamlarının minimum ve maximum değerini bulunuz. (İpucu: Lagrange yöntemi kullanınız.)

Question 7 (10 points) Find the maximum and minimum values of the **sum** of two non-negative numbers x and y which satisfy $x^3 + y^3 + 3x^2y = 15$.

That is: find Max/min of $f(x,y) = x+y$

subject to $g(x,y) = x^3+y^3+3x^2y-15 = 0$

Use Lagrange Multipliers method: set $\nabla f = \lambda \nabla g$;

$$\boxed{\begin{aligned} f_x &= 1 \overset{1}{=} \lambda(3x^2+6xy) = \lambda g_x \\ f_y &= 1 \overset{2}{=} \lambda(3y^2+3x^2) = \lambda g_y \\ & x^3+y^3+3x^2y \overset{3}{=} 15 \end{aligned}}$$

3 equations, 3 unknown ☺

$$\left.\begin{aligned} 3\lambda x^2 + 6\lambda xy &= 1 \\ 3\lambda x^2 + 3\lambda y^2 &= 1 \end{aligned}\right\} \quad \lambda y^2 = 2\lambda xy \underset{\lambda \neq 0}{\Longrightarrow} y^2 = 2xy$$

$y = 0 \overset{③}{\Rightarrow} x^3 = 15 \Rightarrow x = \sqrt[3]{15}$

$y \neq 0 \Rightarrow y = 2x$

$\Downarrow ③$

$x^3 + 8x^3 + 6x^3 = 15$

$x^3 = 1$

$\Downarrow$

$\boxed{x=1} \Rightarrow \boxed{y=2}$

$f(\sqrt[3]{15}, 0) = \sqrt[3]{15} + 0 = \sqrt[3]{15}$ min

$f(1, 2) = 1 + 2 = 3$ MAX

(9) $D: y = 0,\ y = x$, and $x + y = 2$ doğrularıyla sınırlı bölge olmak üzere $\int_D \int (x + 2y)\, dA$ integralini

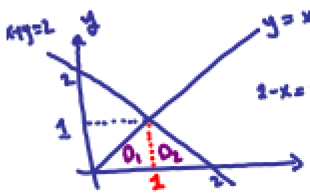
a) $dx\,dy$ sırasında yazınız.

b) $dy\,dx$ sırasında yazınız.

c) $\int_D \int (x + 2y)\, dA$ integralini hesaplayınız.

Question 2 (6+6+8=20 pts) Consider the double integral $\int_D \int (x + 2y)\, dA$ where D is the region in $\mathbf{R}^2$ bounded by the lines $y = 0$, $y = x$, and $x + y = 2$.

a) Write the double integral (do **not** evaluate) as an iterated integral in the order $dxdy$.



$2-x=x \Rightarrow x=1$

$$\iint_D (x+2y)\,dA = \int_0^1 \int_y^{2-y} (x+2y)\,dx\,dy$$

b) Write the double integral (do **not** evaluate) as an iterated integral in the order $dydx$.

$$\iint_{D=D_1\cup D_2} (x+2y)\,dA = \int_0^1 \int_0^x (x+2y)\,dy\,dx + \int_1^2 \int_0^{2-x} (x+2y)\,dy\,dx$$

c) Evaluate $\int_D \int (x + 2y)\, dA$.

From part (a); $\displaystyle\int_0^1 \int_y^{2-y} (x+2y)\,dx\,dy = \int_0^1 \left.\frac{x^2}{2} + 2xy\right|_{x=y}^{x=2-y} dy$

$$= \int_0^1 \frac{(2-y)^2}{2} + 2(2-y)y - \frac{y^2}{2} - 2y^2\,dy = \int_0^1 2 - \underbrace{2y} + \cancel{\frac{y^2}{2}} + \underbrace{4y} - 2y^2 - \cancel{\frac{y^2}{2}} - 2y^2\,dy$$

$$= \int_0^1 (-4y^2 + 2y + 2)\,dy = -\frac{4y^3}{3} + y^2 + 2y\Big|_0^1 = -\frac{4}{3} + 1 + 2 = \frac{5}{3}$$

20) a) $f(x,y) = \frac{1}{3}x^3 - \frac{1}{4}x + y^2$. fonksiyonunun kritik noktalarını bulunuz.

Question 4 (9+15=24 pts)

a) Find and classify the critical points of the function $f(x,y) = \frac{1}{3}x^3 - \frac{1}{4}x + y^2$.

(x_0, y_0) is a critical point of $f(x,y)$ if $f_x(x_0,y_0) = 0 = f_y(x_0,y_0)$.

$$\left.\begin{aligned} f_x(x,y) &= x^2 - \tfrac{1}{4} = 0 \Rightarrow x = \mp 1/2 \\ f_y(x,y) &= 2y = 0 \Rightarrow y = 0 \end{aligned}\right\} \begin{aligned} &(x_1,y_1) = (-1/2, 0) \ \& \ (x_2,y_2) = (1/2, 0) \\ &\text{are the only critical points.} \end{aligned}$$

$$\left.\begin{aligned} f_{xx}(x,y) &= 2x = A \\ f_{yy}(x,y) &= 2 = C \\ f_{xy}(x,y) &= 0 = B \end{aligned}\right\} \Rightarrow \Delta = B^2 - AC = 0 - 4x = -4x$$

- at $(-\frac{1}{2}, 0)$ $\quad \Delta = -4 \cdot (-\frac{1}{2}) = 2 > 0 \Rightarrow (\frac{-1}{2}, 0)$ saddle point
- at $(1/2, 0)$ $\quad \Delta = -4 \ (\frac{1}{2}) = -2 < 0 \ \& \ A = 2 \cdot \frac{1}{2} = 1 > 0 \Rightarrow (\frac{1}{2}, 0)$ is a local minimum point

b) f fonksiyonunun $D=\{(x,y)\in \mathbb{R}^2 : (x-1)^2+4y^2\le 4\}$ bölgesi üzerindeki maksimum ve minimum değerini hesaplayınız. Sınırda Lagrange yöntemi kullanınız.

b) Using Lagrange Multiplier method on the boundary, find the absolute maximum and minimum values of $f(x,y)$ given in part (a) on the region

$$D=\{(x,y)\in \mathbb{R}^2 : (x-1)^2+4y^2\le 4\}.$$

We have checked the critical points inside D in part (a).

Now let's check the boundary points of D $\left(\partial D=\{(x,y)\in\mathbb{R}^2 : (x-1)^2+4y^2=4\}\right)$

ie, Find Max/min of $f(x,y)=\frac{1}{3}x^3-\frac{1}{4}x+y^2$ subject to $g(x,y)=(x-1)^2+4y^2-4=0$] use Lagrange multiplier method.

We need to solve $\nabla f=\lambda\nabla g$:

$$\left.\begin{aligned} x^2-\tfrac{1}{4} &\overset{1}{=} 2\lambda(x-1)\\ 2y &\overset{2}{=} 8\lambda y\\ (x-1)^2+4y^2-4 &\overset{3}{=} 0 \end{aligned}\right]$$

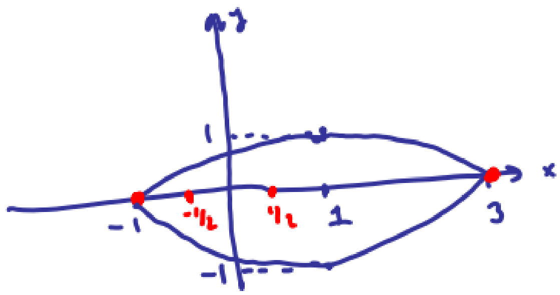
3 unknowns (x,y,λ), 3 equations

$y=0 \overset{3}{\Rightarrow} (x-1)^2=4 \Rightarrow x-1=\mp 2$: $x_1=-1 \Rightarrow (-1,0)$; $x_2=3 \Rightarrow (3,0)$

$y\neq 0 \overset{2}{\Rightarrow} 2=8\lambda \Rightarrow \lambda=\frac{1}{4} \overset{1}{\Rightarrow} x^2-\frac{1}{4}-\frac{1}{2}x+\frac{1}{2}=0 \Rightarrow x^2-\frac{1}{2}x+\frac{1}{4}=0 \Rightarrow$

$4x^2-2x+1=0$

$\Delta=4-4\cdot4\cdot1<0 \Rightarrow$ no solution



Note that since f is a continuous function on the closed domain D, it has absolute max & absolute min on D.

$f(\frac{1}{2},0)=\frac{-1}{12}$ } absolute minimum

$f(-1,0)=\frac{-1}{12}$

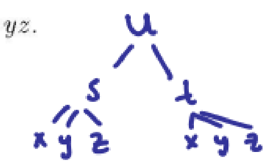
$f(3,0)=\frac{33}{4}$ } absolute maximum

21) $U = F(s,t)$ kısmi türevlere sahip bir fonksiyon ve $s(x,y,z) = xy$ ve $t(x,y,z) = yz$ olmak üzere $x\frac{\partial U}{\partial x} - y\frac{\partial U}{\partial y} + z\frac{\partial U}{\partial z} = 0.$ olduğunu gösteriniz.

Question 5 (9+5=14 pts)

Let $U = F(s,t)$ have partial derivatives where $s(x,y,z) = xy$ and $t(x,y,z) = yz$.

a)Show that $x\frac{\partial U}{\partial x} - y\frac{\partial U}{\partial y} + z\frac{\partial U}{\partial z} = 0.$



$U_x = U_s \cdot S_x + U_t \cdot t_x = U_s \cdot y + U_t \cdot 0$

$U_y = U_s \cdot S_y + U_t \cdot t_y = U_s \cdot x + U_t z$

$U_z = U_s \cdot S_z + U_t \cdot t_z = U_s \cdot 0 + U_t \cdot y$

Hence

$x U_x - y U_y + z U_z = xy U_s - xy U_s - yz U_t + yz U_t = 0$